Momentum and Adaptive Gradient Methods

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Outline

1. AdaGrad
2. RMSProp
3. Momentum
4. Adam

Recall: Gradient descent method

$$\min f(x) = \sum_{i=1}^{n} f_i(x). \tag{1}$$

Gradient descent method:

$$x^{(t+1)} = x^{(t)} - \alpha \nabla f(x^{(t)}) \tag{2}$$

Challenges:

- Slow convergence
- Same step-size for all dimensions, *e.g.*, $f(x) = x_1^2 + 20x_2^2$.
- Noisy gradient
- Non convexity

AdaGrad

$$x^{(t+1)} = x^{(t)} - \alpha \nabla f(x^{(t)}) \oslash G^{(t)}, \tag{3}$$

where $(G^{(t)})_{ii} = \sqrt{\sum_{j=1}^{t} \nabla f(x_j)_i^2}$, $(G^{(t)})_{ii} = 0$ if $i \neq j$.

- A cheap approximation of the Hessian: only estimate the diagonal elements
- Enforces a separate step size / learning rate for every coordinate of x vector
  $\rightarrow$ Decreases the step size for coordinates with high gradient magnitudes
- Potential issue: step size can become very small in some directions

RMSProp

$$x^{(t+1)} = x^{(t)} - \alpha \nabla f(x^{(t)}) \oslash G^{(t)}, \tag{4}$$

where $(G^{(t)})_{ii} = \sqrt{(g^{(t)})_i}$, $(G^{(t)})_{ij} = 0$ if $i \neq j$, and $(g^{(t)})_i = \beta(g^{(t-1)})_i + (1-\beta)(\nabla f(x^{(t)})_i)^2$

- Take a moving average of the coordinates of the gradient, instead of a sum.
- Puts more of an emphasis on the current gradients than the gradient history.
- Can be more effective for DNNs.

Polyak momentum

$$x^{(t+1)} = x^{(t)} - \alpha \nabla f(x^{(t)}) + \beta(x^{(t)} - x^{(t-1)}). \tag{5}$$

- Intuition: if current gradient step is in same direction as previous step, then move a little further in that direction; if not, move less far.
- Also known as the heavy ball method.

Nesterov accelerated gradient

$$x^{(t+1)} = x^{(t)} - \alpha \nabla f \left(x^{(t)} + \beta (x^{(t)} - x^{(t-1)}) \right) + \beta (x^{(t)} - x^{(t-1)}). \quad (6)$$

- A "look-ahead" variant of momentum.
- Provably faster convergence than GD for any convex function.

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Adam

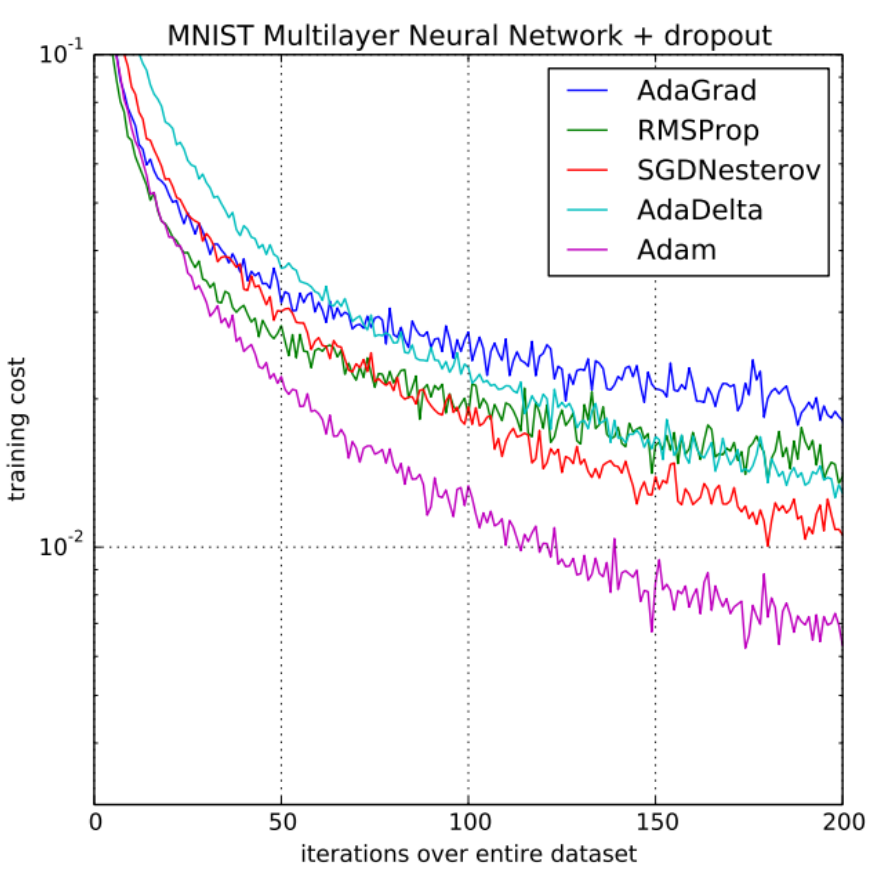
$$x^{(t+1)} = x^{(t)} - \alpha \nabla v^{(t)} \oslash G^{(t)}, \tag{7}$$

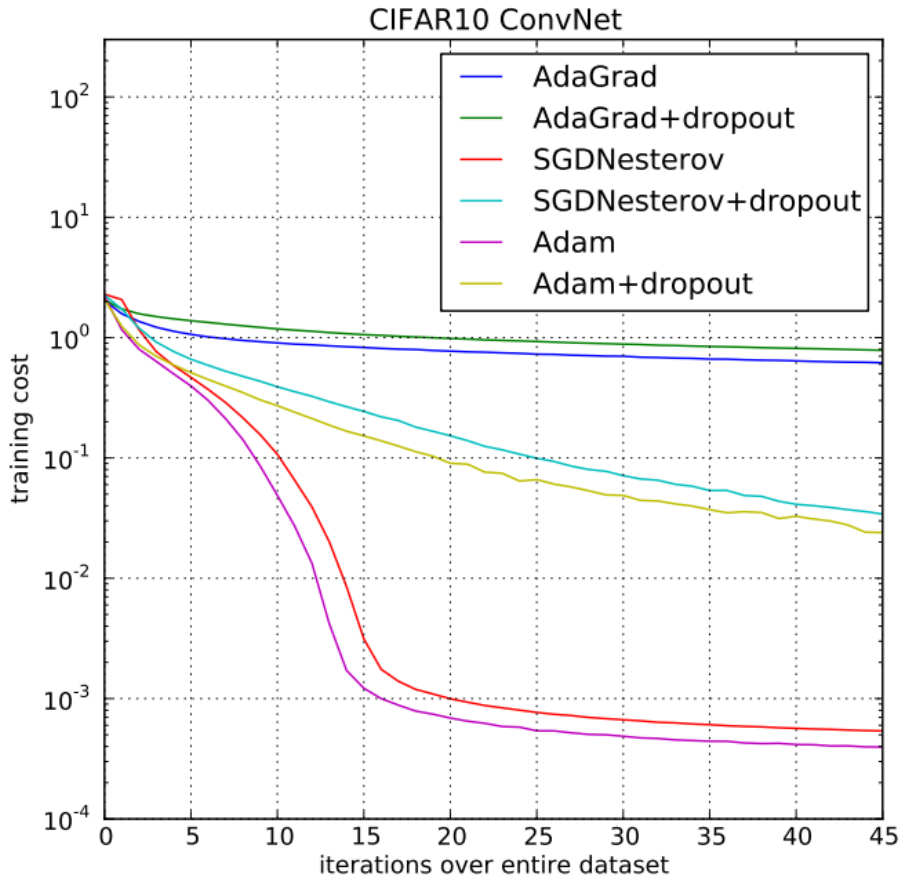
where

$$(G^{(t)})_{ii} = \sqrt{(g^{(t)})_i}, \quad (G^{(t)})_{ii} = 0 \text{ if } \mathrm{i} \neq \mathrm{j},$$
$$(g^{(t)})_i = \beta_1 (g^{(t-1)})_i + (1-\beta_1)(\nabla f(x^{(t)})_i)^2 \leftarrow \text{RMSProp}$$
$$v^{(t)} = \beta_2 v^{(t-1)} + (1-\beta_2)\nabla f(x^{(t)}) \leftarrow \text{momentum}$$

- Combines momentum and RMSprop.
- Default choice in most DL frameworks.

Adam





- Kingma Ba "Adam: A Method for Stochastic Optimization", ICLR 2015.
- `imgur.com/a/visualizing-optimization-algos-Hqolp`

References

Dive into Deep Learning, Chapter 12 (`https://d2l.ai/`)